

Taylor Series

An **infinitely differentiable function** can be expressed as a **Taylor series** expanded around a point x_0 .

$$f(x) = f(x_0) + f'(x_0) \cdot (x - x_0) + \frac{1}{2!} \cdot f''(x_0) \cdot (x - x_0)^2 + \frac{1}{3!} \cdot f^{(3)}(x_0) \cdot (x - x_0)^3 + \dots$$
$$\sum_{n=0}^{\infty} \frac{1}{n!} \cdot f^{(n)}(x_0) \cdot (x - x_0)^n$$

The series expanded around $x_0 = 0$ is called **Maclaurin series**.

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^{2n+1}}{(2n+1)!}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^{2n}}{(2n)!}$$

Taylor Series

```
File Edit Format Run Options Window Help
1  ### Calculations of Sine Function from Taylor Series
2
3  import math
4
5  x = 6.2
6  N = 20
7
8  def sin(x,n):
9      k = int(x/math.pi)
10     x = math.fmod(x,math.pi)
11     f = 0
12     for i in range(n+1):
13         f += ((-1.)**(i))*(x**(2*i+1))/math.factorial(2*i+1)
14
15     if (k%2 == 0): return f
16     else          : return -f
17
18 print(' Calculations of Sine Function from Taylor Series: x =', '%3s' %x)
19 print('%3s' %'n', '%20s' %'Taylor Series', '%20s' %'sin(x)')
20 for n in range(N+1):
21     print('%3s' %n, '%20.16f' %sin(x,n), '%20.16f' %math.sin(x))
22
```

Taylor Series

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```
1 ### Calculations of Sine Function from Taylor Series
```

```
2
3 import math
```

```
4
5 x = 6.2
6 N = 20
```

} ① Input

```
7
8 def sin(x,n):
9     k = int(x/math.pi)
10    x = math.fmod(x,math.pi)
11    f = 0
12    for i in range(n+1):
13        f += ((-1.)**(i))*(x**(2*i+1))/math.factorial(2*i+1)
14
15    if (k%2 == 0): return f
16    else          : return -f
```

} ② Define

```
17
18 print(' Calculations of Sine Function from Taylor Series: x =', '%3s' %x)
19 print('%3s' %'n', '%20s' %'Taylor Series', '%20s' %'sin(x)')
20 for n in range(N+1):
21     print('%3s' %n, '%20.16f' %sin(x,n), '%20.16f' %math.sin(x))
22
```

} ③ Execute

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20 for n in range(N+1):
21     print('%3s' %n, '%20.16f' %sin(x,n), '%20.16f' %math.sin(x))
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starting from $i = 0$

$\text{range}(0, n + 1, 1)$

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20 for n in range(N+1):
21     print('%3s' %n, '%20.16f' %sin(x,n), '%20.16f' %math.sin(x))
22
```

stop immediately when $i = n + 1$
 $\text{range}(0, n + 1, 1)$

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20 for n in range(N+1):
21     print('%3s' %n, '%20.16f' %sin(x,n), '%20.16f' %math.sin(x))
22
```

stop immediately when $i = n + 1$ (do not perform calculation for $i = n + 1$)

$\text{range}(0, n + 1, 1)$

Taylor Series

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20 for n in range(N+1):
21     print('%3s' %n, '%20.16f' %sin(x,n), '%20.16f' %math.sin(x))
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```

increasing by 1

range(0, n + 1, 1)

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20 for n in range(N+1):
21     print('%3s' %n, '%20.16f' %sin(x,n), '%20.16f' %math.sin(x))
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```

 $i = 0$ x

Taylor Series

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```

$$i = 1$$
$$x - \frac{x^3}{3!}$$

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```

$$i = 2$$
$$x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

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7
8  def sin(x,n):
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20 for n in range(N+1):
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22
```

 $i = 3$

$$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$$

Taylor Series

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```

$$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Taylor Series

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5  x = 6.2
6  N = 20
7
8  def sin(x,n):
9      k = int(x/math.pi)
10     x = math.fmod(x,math.pi)
11     f = 0
12     for i in range(n+1):
13         f += ((-1.)**(i))*(x**(2*i+1))/math.factorial(2*i+1)
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15     if (k%2 == 0): return f
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18 print(' Calculations of Sine Function from Taylor Series: x =', '%3s' %x)
19 print('%3s' %'n', '%20s' %'Taylor Series', '%20s' %'sin(x)')
20 for n in range(N+1):
21     print('%3s' %n, '%20.16f' %sin(x,n), '%20.16f' %math.sin(x))
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```

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```

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15     if (k%2 == 0): return f
16     else           : return -f
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18 print(' Calculations of Sine Function from Taylor Series: x =', '%3s' %x)
19 print('%3s' %'n', '%20s' %'Taylor Series', '%20s' %'sin(x)')
20 for n in range(N+1):
21     print('%3s' %n, '%20.16f' %sin(x,n), '%20.16f' %math.sin(x))
22

```

Calculations of Sine Function from Taylor Series: x = 6.2

n	Taylor Series	sin(x)
0	-3.0584073464102071	-0.0830894028174964
1	1.7095760482543430	-0.0830894028174964
2	-0.5203753359339442	-0.0830894028174964
3	-0.0237409785033086	-0.0830894028174964
4	-0.0882610620316559	-0.0830894028174964
5	-0.0827745935048270	-0.0830894028174964
6	-0.0831035655162788	-0.0830894028174964
7	-0.0830889123893381	-0.0830894028174964
8	-0.0830894162982792	-0.0830894028174964
9	-0.0830894025161406	-0.0830894028174964
10	-0.0830894028230837	-0.0830894028174964
11	-0.0830894028174096	-0.0830894028174964
12	-0.0830894028174981	-0.0830894028174964
13	-0.0830894028174969	-0.0830894028174964
14	-0.0830894028174969	-0.0830894028174964
15	-0.0830894028174969	-0.0830894028174964
16	-0.0830894028174969	-0.0830894028174964
17	-0.0830894028174969	-0.0830894028174964
18	-0.0830894028174969	-0.0830894028174964
19	-0.0830894028174969	-0.0830894028174964
20	-0.0830894028174969	-0.0830894028174964

Taylor Series

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11     f = 0
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Calculations of Sine Function from Taylor Series: x = 6.2

n	Taylor Series	sin(x)
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1	1.7095760482543430	-0.0830894028174964
2	-0.5203753359339442	-0.0830894028174964
3	-0.0237409785033086	-0.0830894028174964
4	-0.0882610620316559	-0.0830894028174964
5	-0.0827745935048270	-0.0830894028174964
6	-0.0831035655162788	-0.0830894028174964
7	-0.0830889123893381	-0.0830894028174964
8	-0.0830894162982792	-0.0830894028174964
9	-0.0830894025161406	-0.0830894028174964
10	-0.0830894028230837	-0.0830894028174964
11	-0.0830894028174096	-0.0830894028174964
12	-0.0830894028174981	-0.0830894028174964
13	-0.0830894028174969	-0.0830894028174964
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15	-0.0830894028174969	-0.0830894028174964
16	-0.0830894028174969	-0.0830894028174964
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19	-0.0830894028174969	-0.0830894028174964
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10     x = math.fmod(x,math.pi)
11     f = 1.
12     for i in range(n,-1,-1):
13         if (i != 0): f = 1. - f*x/(2*i+1)*x/(2*i)
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Calculations of Sine Function from Taylor Series: x = 6.2

n	Taylor Series	sin(x)
0	-3.0584073464102071	-0.0830894028174964
1	1.7095760482543430	-0.0830894028174964
2	-0.5203753359339438	-0.0830894028174964
3	-0.0237409785033084	-0.0830894028174964
4	-0.0882610620316548	-0.0830894028174964
5	-0.0827745935048267	-0.0830894028174964
6	-0.0831035655162778	-0.0830894028174964
7	-0.0830889123893378	-0.0830894028174964
8	-0.0830894162982791	-0.0830894028174964
9	-0.0830894025161400	-0.0830894028174964
10	-0.0830894028230829	-0.0830894028174964
11	-0.0830894028174097	-0.0830894028174964
12	-0.0830894028174980	-0.0830894028174964
13	-0.0830894028174960	-0.0830894028174964
14	-0.0830894028174960	-0.0830894028174964
15	-0.0830894028174960	-0.0830894028174964
16	-0.0830894028174960	-0.0830894028174964
17	-0.0830894028174960	-0.0830894028174964
18	-0.0830894028174960	-0.0830894028174964
19	-0.0830894028174960	-0.0830894028174964
20	-0.0830894028174960	-0.0830894028174964

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8	-0.0830894162982791	-0.0830894028174964
9	-0.0830894025161400	-0.0830894028174964
10	-0.0830894028230829	-0.0830894028174964
11	-0.0830894028174097	-0.0830894028174964
12	-0.0830894028174980	-0.0830894028174964
13	-0.0830894028174960	-0.0830894028174964
14	-0.0830894028174960	-0.0830894028174964
15	-0.0830894028174960	-0.0830894028174964
16	-0.0830894028174960	-0.0830894028174964
17	-0.0830894028174960	-0.0830894028174964
18	-0.0830894028174960	-0.0830894028174964
19	-0.0830894028174960	-0.0830894028174964
20	-0.0830894028174960	-0.0830894028174964

Taylor's Theorem

A function that can be expanded around a point x_0 may be expressed in the following form.

$$f(x) = \sum_{k=0}^{\infty} \frac{1}{k!} \cdot f^{(k)}(x_0) \cdot (x - x_0)^k$$

$$= \sum_{k=0}^n \frac{1}{k!} \cdot f^{(k)}(x_0) \cdot (x - x_0)^k + \frac{1}{(n+1)!} \cdot f^{(n+1)}(\xi) \cdot (x - x_0)^{n+1}; \quad x_0 < \xi < x$$

Taylor Polynomials: $P_n(x) = \sum_{k=0}^n \frac{1}{k!} \cdot f^{(k)}(x_0) \cdot (x - x_0)^k$

Remainders: $R_n(x) = \frac{1}{(n+1)!} \cdot f^{(n+1)}(\xi) \cdot (x - x_0)^{n+1}; \quad x_0 < \xi < x$

The function $f(x)$ can be approximated using Taylor polynomials.

$$\cos 0.01 = 0.999950000416665 \dots \quad \begin{matrix} x_0 = 0 \\ x = 0.01 \end{matrix} \rightarrow \begin{cases} \cos 0.01 \approx P_2(0.01) = 1 - \frac{(0.01)^2}{2!} = 0.999950000000000 \\ \cos 0.01 \approx P_4(0.01) = 1 - \frac{(0.01)^2}{2!} + \frac{(0.01)^4}{4!} = 0.999950000416666 \dots \end{cases}$$